

AtmoSync - Real-Time IoT Monitoring Dashboard

Introduction

AtmoSync is a real-time IoT monitoring system that simulates sensor data (temperature, humidity, vibration), streams it using Apache Kafka, stores it in SQLite, and visualizes it through a Streamlit dashboard.

Objective

Generate live sensor data, stream it through Kafka, store it in a database, and display it on a dashboard to demonstrate a real-time data pipeline.

Technologies Used

Python, Apache Kafka, ZooKeeper, Docker, SQLite, Streamlit, Plotly, Pandas.

Architecture

Sensor Simulator → Producer → Kafka → Consumer → SQLite Database → Streamlit Dashboard.

Workflow

1. Producer generates random sensor values every 2 seconds.
2. Producer sends JSON messages to the Kafka topic container_sensor_data.
3. Kafka temporarily stores messages and delivers them to the consumer.
4. Consumer reads messages and stores them in SQLite.
5. Streamlit reads the database and displays live tables and charts.

Roles of Components

- Producer: Generates and sends sensor data.
- Kafka: High-speed message broker between producer and consumer.
- ZooKeeper: Coordinates Kafka services.
- Consumer: Receives Kafka messages and stores them.
- SQLite: Stores sensor records.
- Streamlit: Displays real-time dashboard.

Applications

Cold chain monitoring, logistics, smart warehouses, manufacturing, agriculture, vaccine transport.

Future Enhancements

Real IoT devices, cloud database, authentication, alerts, machine learning, cloud deployment.

Presentation Summary

AtmoSync demonstrates an end-to-end real-time IoT data pipeline using modern technologies. It simulates sensors, streams data through Kafka, stores it in SQLite, and visualizes it live using Streamlit.